Statistical and Machine Learning

Regularisation and Bayesian Methods

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Exercise 1 – From OLS to Regularisation

The penalised least squares (PLS) criterion is given by

$$PLS(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda\, pen(\boldsymbol{\beta})$$

a) Explain why OLS can fail when predictors are highly collinear or when $p > n$. What goes wrong algebraically?

Model answer: The OLS estimator requires computing $(\mathbf{X}^\top\mathbf{X})^{-1}$. When predictors are highly collinear, $\mathbf{X}^\top\mathbf{X}$ becomes nearly singular: its determinant is close to zero and the inverse is numerically unstable, producing very large and erratic coefficient estimates with huge variances. When $p > n$, $\mathbf{X}^\top\mathbf{X}$ is a $p \times p$ matrix of rank at most $n < p$, so it is singular and the inverse does not exist at all. In this regime, the system $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}$ is underdetermined and has infinitely many solutions, all achieving $RSS = 0$ on the training data.

b) The PLS problem can equivalently be written as a constrained optimisation:

$$\hat{\boldsymbol{\beta}}_{PLS} = \arg\min_{\boldsymbol{\beta}} \, (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \quad \text{subject to} \quad pen(\boldsymbol{\beta}) \leq t$$

Explain in words what role the constraint $pen(\boldsymbol{\beta}) \leq t$ plays. What happens to $\hat{\boldsymbol{\beta}}_{PLS}$ as $t \to \infty$?

Model answer: The constraint restricts the solution to lie within a region of coefficient space defined by $pen(\boldsymbol{\beta}) \leq t$ — a ball of radius t under the chosen penalty norm. This rules out the large-coefficient solutions that OLS would otherwise find in collinear or $p > n$ settings, effectively selecting a unique, stable solution among the infinitely many that minimise the residual sum of squares. As $t \to \infty$ the constraint becomes non-binding (every $\boldsymbol{\beta}$ satisfies it), so the constrained problem reduces to unconstrained OLS and $\hat{\boldsymbol{\beta}}_{PLS} \to \hat{\boldsymbol{\beta}}_{LS}$.

c) The tuning parameter λ controls the strength of the penalty. Describe in words what happens to $\hat{\boldsymbol{\beta}}_{PLS}$ as $\lambda \to 0$ and as $\lambda \to \infty$.

Model answer: As $\lambda \to 0$, the penalty term vanishes and the criterion reduces to the OLS objective, so $\hat{\boldsymbol{\beta}}_{PLS} \to \hat{\boldsymbol{\beta}}_{LS}$. As $\lambda \to \infty$, the penalty dominates and any non-zero coefficient incurs an arbitrarily large cost, forcing all coefficients toward zero: $\hat{\boldsymbol{\beta}}_{PLS} \to \mathbf{0}$. For intermediate λ, the solution is a compromise between fitting the data and keeping coefficients small — i.e., the coefficients are shrunk toward zero relative to OLS.

Exercise 2 — Ridge vs. Lasso

Ridge regression uses $pen(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_2^2 = \sum_{k=1}^{p} \beta_k^2$ and Lasso uses $pen(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_1 = \sum_{k=1}^{p} |\beta_k|$.

a) For orthonormal covariates ($\mathbf{X}^\top\mathbf{X} = \mathbf{I}_p$), the ridge and Lasso estimators for each coefficient j reduce to simple expressions. Fill in the table below and comment on what each operation does to $\hat{\beta}_{j,LS}$.

Method	$\hat{\beta}_j(\lambda)$	Operation on $\hat{\beta}_{j,LS}$
Ridge		
Lasso		

Model answer:

Method	$\hat{\beta}_j(\lambda)$	Operation on $\hat{\beta}_{j,LS}$
Ridge	$\dfrac{1}{1+\lambda}\hat{\beta}_{j,LS}$	Proportional shrinkage: scales every coefficient by the same factor $\frac{1}{1+\lambda} < 1$. No coefficient is set exactly to zero.
Lasso	$\text{sign}(\hat{\beta}_{j,LS}) \left(\lvert\hat{\beta}_{j,LS}\rvert - \frac{\lambda}{2}\right)_+$	Soft thresholding: subtracts $\frac{\lambda}{2}$ from the absolute value and clips at zero. Coefficients smaller than $\frac{\lambda}{2}$ in magnitude are set exactly to zero.

b) The Lasso objective is not differentiable everywhere. At which points does differentiability fail, and what does this imply for how the solution must be found?

Model answer: The ℓ_1 penalty $\sum_k |\beta_k|$ is not differentiable at $\beta_k = 0$ for any k: the function has a kink there. Consequently, standard gradient-based methods (which require a smooth gradient) cannot be applied directly. The solution must be found using subgradient methods, coordinate descent, or specialised algorithms such as LARS (Least Angle Regression). In R, `glmnet` uses coordinate descent.

c) Ridge regression has a closed-form solution whereas Lasso does not. State the closed-form ridge estimator. Why does adding $\lambda\mathbf{I}_p$ to $\mathbf{X}^\top\mathbf{X}$ solve the invertibility problem that arises in OLS when $p > n$?

Model answer: The ridge estimator is

$$\hat{\boldsymbol{\beta}}_{PLS} = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\mathbf{X}^\top\mathbf{y}$$

For any $\lambda > 0$, the matrix $\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}_p$ is strictly positive definite (all eigenvalues exceed $\lambda > 0$), hence invertible, regardless of whether $\mathbf{X}^\top\mathbf{X}$ is singular or not. Adding $\lambda\mathbf{I}_p$ shifts all eigenvalues of $\mathbf{X}^\top\mathbf{X}$ upward by λ, eliminating any zero eigenvalues and stabilising the inversion.

d) Consider the constraint-region picture (diamond for Lasso, circle for Ridge). What do the contours refer to? What do the shaded areas refer to? Explain geometrically why Lasso tends to produce exactly zero coefficients while Ridge does not. Explain terms *shrinkage* and *sparsity* in the context of ridge regression and LASSO, respectively.

Model answer: The constrained optimum lies where a contour of the least-squares objective function first touches the boundary of the constraint region. The ridge constraint region is a circle (in 2D), which has a smooth, curved boundary with no special points. The contour is unlikely to touch it exactly on an axis, so neither coefficient is exactly zero. The Lasso constraint region is a diamond with corners located on the coordinate axes. Because contours of the least-squares objective are ellipses, they are likely to first contact the diamond at a corner — a geometrically prominent point — setting one or more coefficients to exactly zero. This is the geometric origin of Lasso's variable selection property.

Exercise 3 — Bias, Variance and the Choice of λ

a) Under standard linear model assumptions, OLS is unbiased: $\mathbb{E}(\hat{\boldsymbol{\beta}}_{LS}) = \boldsymbol{\beta}$. Show that the ridge estimator is biased by writing down $\mathbb{E}(\hat{\boldsymbol{\beta}}_{PLS})$ and explaining why it differs from $\boldsymbol{\beta}$.

Model answer: Let $\mathbf{A}_\lambda = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\mathbf{X}^\top\mathbf{X}$. Then

$$\mathbb{E}(\hat{\boldsymbol{\beta}}_{PLS}) = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\mathbf{X}^\top\mathbf{X}\,\boldsymbol{\beta} = \mathbf{A}_\lambda\,\boldsymbol{\beta}$$

Since $\mathbf{A}_\lambda \neq \mathbf{I}_p$ for any $\lambda > 0$ (the ridge penalty shrinks the eigenvalues of $\mathbf{X}^\top\mathbf{X}$), we have $\mathbb{E}(\hat{\boldsymbol{\beta}}_{PLS}) \neq \boldsymbol{\beta}$ in general. The bias is $(\mathbf{A}_\lambda - \mathbf{I}_p)\boldsymbol{\beta} = -\lambda(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\boldsymbol{\beta}$, which grows in magnitude as λ increases.

b) It can be shown that $Var(\hat{\boldsymbol{\beta}}_{PLS}) \leq Var(\hat{\boldsymbol{\beta}}_{LS})$ in the matrix sense. Explain intuitively why introducing the penalty reduces variance, and what is the price paid for this reduction.

Model answer: Variance in $\hat{\boldsymbol{\beta}}_{LS}$ comes from the sensitivity of $(\mathbf{X}^\top\mathbf{X})^{-1}$ to small perturbations in the data, especially when $\mathbf{X}^\top\mathbf{X}$ is near-singular. By adding $\lambda\mathbf{I}_p$, ridge regression effectively regularises this inversion: the matrix being inverted is better conditioned (eigenvalues are bounded away from zero), so small changes in $\mathbf{y}$ lead to smaller changes in $\hat{\boldsymbol{\beta}}_{PLS}$. The price is bias — the estimator is systematically pulled toward zero, so it no longer targets the true $\boldsymbol{\beta}$ on average.

c) Recall that $MSE = Bias^2 + Var$. Sketch (or describe) how the bias and variance of $\hat{\boldsymbol{\beta}}_{PLS}$ each change as λ increases from 0 to ∞. What shape does the MSE curve take, and what does this imply for choosing λ?

Model answer: As λ increases from 0:

- Bias increases monotonically from 0 (at $\lambda = 0$, ridge reduces to OLS which is unbiased) toward a maximum as coefficients are pushed further from their true values.
- Variance decreases monotonically from $Var(\hat{\boldsymbol{\beta}}_{LS})$ toward 0 (at $\lambda \to \infty$ the estimator is identically $\mathbf{0}$, which has zero variance).
- MSE = Bias^2 + Variance is U-shaped: it starts at the OLS MSE, decreases initially as the variance reduction outweighs the bias increase, reaches a minimum at some $\lambda^* > 0$, and then increases as bias dominates. This implies that there is always an optimal $\lambda^* > 0$ that achieves lower MSE than OLS, and cross-validation is used to find it in practice.

d) Cross-validation is the standard method for selecting λ. Briefly describe the K-fold CV procedure applied to ridge regression. What quantity is minimised, and over what?

Model answer: For a grid of candidate values $\lambda_1, \ldots, \lambda_m$:

1. Randomly partition the data into K folds of approximately equal size.
2. For each fold $k = 1, \ldots, K$: fit the ridge estimator $\hat{\boldsymbol{\beta}}_{PLS,(-k)}(\lambda)$ on the remaining $K-1$ folds and compute the prediction error on fold k:

$$CV(\lambda)_k = \frac{1}{|D_k|}\sum_{i \in D_k}(y_i - \mathbf{x}_i^\top\hat{\boldsymbol{\beta}}_{PLS,(-k)}(\lambda))^2$$

3. Average over folds: $CV(\lambda) = \frac{1}{K}\sum_{k=1}^K CV(\lambda)_k$.
4. Select $\hat{\lambda} = \arg\min_\lambda CV(\lambda)$.

The quantity minimised is the cross-validated mean squared prediction error $CV(\lambda)$, and the optimisation is over the tuning parameter λ.

Exercise 4 — Extensions of Regularisation: Elastic Net and Group Lasso

a) Suppose two predictors are perfectly correlated: $X_1 = X_2$. Describe what the Ridge and Lasso estimators each do in this case. Which method behaves more naturally, and why?

Model answer: When $X_1 = X_2$, the two predictors are indistinguishable from the data. Ridge regression always produces $\hat{\beta}_1 = \hat{\beta}_2$: by symmetry it splits the effect equally between them, which is natural since the variables are identical. The Lasso solution in this case is not unique – any combination $(\hat{\beta}_1, \hat{\beta}_2)$ satisfying $\hat{\beta}_1 + \hat{\beta}_2 = c$ (for some c) minimises the objective equally well, because only the sum enters the ℓ_1 penalty and the predictions. This non-uniqueness is problematic in practice: the Lasso may arbitrarily select one variable and drop the other, an unstable choice with no principled justification. Ridge therefore behaves more naturally here by treating the correlated variables symmetrically.

b) The Elastic Net combines ℓ_1 and ℓ_2 penalties with a mixing parameter $\alpha \in [0, 1]$:

$$pen(\boldsymbol{\beta}) = \sum_{j=1}^{p} \left(\alpha|\beta_j| + (1-\alpha)\beta_j^2\right)$$

What does the Elastic Net reduce to when $\alpha = 1$? When $\alpha = 0$? Explain what intermediate values of α achieve that neither Lasso nor Ridge alone can.

Model answer: When $\alpha = 1$ the ℓ_2 term vanishes and Elastic Net reduces to the Lasso. When $\alpha = 0$ the ℓ_1 term vanishes and it reduces to Ridge regression. For intermediate $\alpha \in (0, 1)$, the Elastic Net combines the strengths of both: the ℓ_1 component encourages sparsity and variable selection, while the ℓ_2 component handles groups of correlated variables by distributing coefficients more evenly among them and producing a unique solution. This is exactly what neither pure Lasso (non-unique, unstable for correlated predictors) nor pure Ridge (no variable selection, never sets coefficients to zero) achieves alone.

c) The Group Lasso penalty is $\sum_{g=1}^{G} \|\boldsymbol{\beta}_{I_g}\|_2$. Explain how this differs from standard ℓ_2 regularisation, and give a concrete example of when it would be more appropriate than the ordinary Lasso.

Model answer: Standard ℓ_2 regularisation penalises $\|\boldsymbol{\beta}\|_2^2 = \sum_j \beta_j^2$, which shrinks all coefficients toward zero but never sets any to exactly zero. The Group Lasso instead applies an ℓ_2 norm within each group and sums these across groups: $\sum_g \|\boldsymbol{\beta}_{I_g}\|_2$. This is a sum of norms, not a sum of squares — and it induces sparsity at the group level rather than at the individual coefficient level. The entire group is either included (all non-zero) or excluded (all exactly zero) together. A concrete example: suppose X_1 is a continuous predictor and $X_2 = X_1^2$ is its square (a polynomial basis expansion). Including X_2 in the model without X_1 would be difficult to interpret. The Group Lasso with $I_1 = \{1, 2\}$ ensures both enter or leave the model together, enforcing meaningful model structure. Similarly useful for categorical variables encoded as multiple dummy variables, which should naturally be selected as a group.

Exercise 5 — Bayesian Inference and Bayes' Formula

a) In the frequentist framework, μ and σ^2 in $X \sim N(\mu, \sigma^2)$ are treated as fixed unknown parameters. How does the Bayesian framework differ in its treatment of these quantities? What does Bayes' theorem then allow us to compute?

Model answer: In the frequentist framework, μ and σ^2 are fixed but unknown constants, and the goal is to estimate them from data — there is no probability distribution over parameters. In the Bayesian framework, μ and σ^2 are treated as random variables with their own distributions, reflecting our uncertainty about them before observing data. We encode this uncertainty in a prior distribution $p(\mu, \sigma^2)$. Bayes' theorem then allows us to compute the posterior distribution $p(\mu, \sigma^2 \mid x_1, \ldots, x_n)$, which combines the prior belief with the information in the data via the likelihood. The posterior represents our updated belief about the parameters after observing the data.

b) Write down Bayes' theorem in the form used for Bayesian inference and name each term. Why is the normalising constant $p(\mathbf{y})$ often omitted?

Model answer:

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{p(\mathbf{y})} \propto p(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})$$

- $p(\boldsymbol{\theta})$: the **prior** distribution — our belief about $\boldsymbol{\theta}$ before observing data.
- $p(\mathbf{y} \mid \boldsymbol{\theta})$: the **likelihood** — how probable the observed data $\mathbf{y}$ is under parameter $\boldsymbol{\theta}$.
- $p(\boldsymbol{\theta} \mid \mathbf{y})$: the **posterior** distribution — our updated belief about $\boldsymbol{\theta}$ after observing $\mathbf{y}$.

- $p(\mathbf{y}) = \int p(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}$: the **marginal likelihood** or evidence — a normalising constant that ensures the posterior integrates to 1.

The normalising constant $p(\mathbf{y})$ does not depend on $\boldsymbol{\theta}$ and is often difficult or impossible to compute analytically in high dimensions. Since it is just a constant with respect to $\boldsymbol{\theta}$, it does not affect the shape of the posterior or the location of its mode. For many purposes (e.g., finding the posterior mode, or comparing relative probabilities), the unnormalised posterior $p(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})$ suffices.

c) Interpret the Boy Who Cried Wolf update in terms of Bayes' theorem. What does it illustrate about Bayesian learning?

Model answer: The prior is the villagers' initial belief before observing any of the boy's behaviour: $P(\text{honest}) = 0.90$, $P(\text{untrustworthy}) = 0.05$, $P(\text{malicious}) = 0.05$. The likelihood is the probability of the observed event — the boy raising a false alarm — under each hypothesis about his character (taken from the table in the slides). The posterior is the updated belief after the observation: $P(\text{honest} \mid \text{lied once}) = 0.114$, a dramatic drop from 0.90. This illustrates several key features of Bayesian learning: (i) the posterior is a data-driven update of the prior; (ii) each new observation feeds in sequentially — the posterior after one lie becomes the prior before the next; (iii) repeated evidence compounds: after two lies the posterior falls to 0.0017, showing how trust is eroded by accumulating evidence, consistent with the moral of the fable.

d) What is a conjugate prior? State one conjugate prior pair and explain the computational advantage.

Model answer: A prior $p(\boldsymbol{\theta}) \in \mathcal{F}$ is conjugate for a likelihood $p(\mathbf{y} \mid \boldsymbol{\theta})$ if the posterior $p(\boldsymbol{\theta} \mid \mathbf{y})$ belongs to the same parametric family $\mathcal{F}$ as the prior. In other words, the functional form of the distribution is preserved under Bayesian updating — only the parameters of the distribution change. One example: if $X_1, \ldots, X_n \mid \mu \sim N(\mu, \sigma^2)$ (known σ^2) and $\mu \sim N(\mu_{\text{prior}}, \sigma^2_{\text{prior}})$, then the posterior is also normal: $\mu \mid x_1, \ldots, x_n \sim N(\mu_{\text{post}}, \sigma^2_{\text{post}})$. The computational advantage is that we never need to evaluate the intractable normalising constant $p(\mathbf{y})$ or perform numerical integration — the posterior is fully characterised by updating a small number of hyperparameters (μ_{prior} and σ^2_{prior} become μ_{post} and σ^2_{post}) using closed-form formulas.

Exercise 6 — Bayesian Interpretation of Regularisation

a) Show that the posterior mean under the normal prior coincides with the Ridge estimator and identify λ.

Model answer: Under the stated model, $\mathbf{Y} \mid \boldsymbol{\beta} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}_n)$ and $\boldsymbol{\beta} \sim N(\mathbf{0}, \tau^2\mathbf{I}_p)$, the general Bayesian linear regression formula for the posterior mean gives:

$$E(\boldsymbol{\beta} \mid \mathbf{Y}) = \left(\mathbf{X}^\top\boldsymbol{\Sigma}^{-1}\mathbf{X} + \boldsymbol{\Sigma}_0^{-1}\right)^{-1}\mathbf{X}^\top\boldsymbol{\Sigma}^{-1}\mathbf{Y}$$

Substituting $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}_n$ and $\boldsymbol{\Sigma}_0 = \tau^2\mathbf{I}_p$ (so $\boldsymbol{\Sigma}_0^{-1} = \frac{1}{\tau^2}\mathbf{I}_p$ and $\boldsymbol{\beta}_0 = \mathbf{0}$):

$$E(\boldsymbol{\beta} \mid \mathbf{Y}) = \left(\frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X} + \frac{1}{\tau^2}\mathbf{I}_p\right)^{-1}\frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{Y} = \left(\mathbf{X}^\top\mathbf{X} + \frac{\sigma^2}{\tau^2}\mathbf{I}_p\right)^{-1}\mathbf{X}^\top\mathbf{Y}$$

This is exactly the Ridge estimator $\hat{\boldsymbol{\beta}}_{PLS} = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\mathbf{X}^\top\mathbf{Y}$ with $\lambda = \frac{\sigma^2}{\tau^2}$. The penalty parameter is therefore the ratio of error variance to prior variance: large τ^2 (diffuse prior, little prior information) gives small λ (weak penalty, close to OLS); small τ^2 (tight prior near zero) gives large λ (strong shrinkage).

b) Show algebraically that maximising the posterior is equivalent to minimising the Ridge criterion.

Model answer: Taking the log of the unnormalised posterior:

$$\log p(\boldsymbol{\beta} \mid \mathbf{y}) \propto \log p(\mathbf{y} \mid \boldsymbol{\beta}) + \log p(\boldsymbol{\beta})$$

With the stated distributions:

$$\log p(\mathbf{y} \mid \boldsymbol{\beta}) = -\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \text{const}$$

$$\log p(\boldsymbol{\beta}) = -\frac{1}{2\tau^2}\boldsymbol{\beta}^\top\boldsymbol{\beta} + \text{const}$$

Maximising $\log p(\boldsymbol{\beta} \mid \mathbf{y})$ is equivalent to minimising $-2\log p(\boldsymbol{\beta} \mid \mathbf{y})$, which equals (up to constants):

$$\frac{1}{\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \frac{1}{\tau^2}\boldsymbol{\beta}^\top\boldsymbol{\beta}$$

Multiplying through by σ^2 (which does not change the argmin):

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \frac{\sigma^2}{\tau^2}\boldsymbol{\beta}^\top\boldsymbol{\beta}$$

This is precisely the Ridge penalised least squares criterion with $\lambda = \frac{\sigma^2}{\tau^2}$.

Exercise 7 — Conjugate priors

a) Assume $Y \mid \theta \sim \text{Binomial}(n, \theta)$, $0 < \theta < 1$, with $p(y \mid \theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$, $y = 0, 1, \ldots, n$. Let the prior distribution be $\theta \sim \text{Beta}(\alpha, \beta)$, $\alpha > 0$, $\beta > 0$ with density $p(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha\Gamma(\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$, $0 < \theta < 1$. Using the Bayes theorem, prove that the beta distribution is a conjugate prior for the binomial likelihood, i.e., that a beta prior combined with a binomial likelihood produces a beta posterior.

Model answer:

1. Assume $Y \mid \theta \sim \text{Binomial}(n, \theta)$, $0 < \theta < 1$. The probability mass function is $p(y \mid \theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$, $y = 0, 1, \ldots, n$. As a function of θ, the likelihood is therefore $L(\theta; y_1, \ldots, y_n) = \prod_{i=1}^n \binom{n}{y_i}\theta^{y_i}(1-\theta)^{n-y_i}$. Since $\binom{n}{y}$ does not depend on θ, $L(\theta; y) \propto \theta^y(1-\theta)^{n-y}$

2. Let the prior distribution be $\theta \sim \text{Beta}(\alpha, \beta)$,$\alpha > 0$, $\beta > 0$. Its density is $p(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$, $0 < \theta < 1$. The normalizing constant $\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$ does not depend on θ, so $p(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$.

3. By Bayes' theorem, $p(\theta \mid y) = \frac{p(y|\theta)p(\theta)}{p(y)}$. The denominator is $p(y) = \int_0^1 p(y \mid \theta)p(\theta)\, d\theta$. Since $p(y)$ does not depend on θ, $p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$.

4. Substitute the binomial likelihood and beta prior: $p(\theta \mid y) \propto \left[\theta^y(1-\theta)^{n-y}\right]\left[\theta^{\alpha-1}(1-\theta)^{\beta-1}\right]$. Collect terms with the same base: $p(\theta \mid y) \propto \theta^{y+\alpha-1}(1-\theta)^{n-y+\beta-1}$. Equivalently, $p(\theta \mid y) \propto \theta^{\alpha+y-1}(1-\theta)^{\beta+n-y-1}$.

5. Recall that the kernel of a beta distribution is $\theta^{a-1}(1-\theta)^{b-1}$. Comparing, $a - 1 = \alpha + y - 1$ so $a = \alpha + y$. Also, $b - 1 = \beta + n - y - 1$ so $b = \beta + n - y$. Therefore, $\theta \mid y \sim \text{Beta}(\alpha + y, \beta + n - y)$.

6. To verify the full posterior density, include the normalizing constant: $p(\theta \mid y) = \frac{\Gamma(\alpha+\beta+n)}{\Gamma(\alpha+y)\Gamma(\beta+n-y)}\theta^{\alpha+y-1}(1-\theta)^{\beta+n-y-1}$, $0 < \theta < 1$. This is exactly the density of $\text{Beta}(\alpha + y, \beta + n - y)$.

7. Therefore, a beta prior combined with a binomial likelihood produces a beta posterior. Hence, the beta distribution is a conjugate prior for the binomial likelihood.

8. Interpretation:
$$\alpha_{\text{post}} = \alpha + y, \quad \beta_{\text{post}} = \beta + n - y.$$
The number of observed successes y is added to the first prior parameter, and the number of observed failures $n - y$ is added to the second prior parameter.

b) Assume $Y \mid \lambda \sim \text{Poisson}(\lambda)$, $\lambda > 0$. The probability mass function is $p(y \mid \lambda) = \frac{e^{-\lambda}\lambda^y}{y!}$, $y = 0, 1, 2, \ldots$. Let the prior distribution be $\lambda \sim \text{Gamma}(\alpha, \beta)$, $\alpha > 0$, $\beta > 0$, where α is the shape parameter and β is the rate paramete, with density $p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda}$, $\lambda > 0$. Using the Bayes theorem, prove that the gamma distribution is a conjugate prior for the Poisson likelihood, i.e., that a gamma prior combined with a Poisson likelihood produces a gamma posterior.

1. Assume $Y \mid \lambda \sim \text{Poisson}(\lambda)$, $\lambda > 0$. The probability mass function is $p(y \mid \lambda) = \frac{e^{-\lambda}\lambda^y}{y!}$, $y = 0, 1, 2, \ldots$. As a function of λ, the likelihood is $L(\lambda; y) = \frac{e^{-\lambda}\lambda^y}{y!}$. Since $y!$ does not depend on λ, $L(\lambda; y) \propto e^{-\lambda}\lambda^y$.
2. Let the prior distribution be $\lambda \sim \text{Gamma}(\alpha, \beta)$, $\alpha > 0,\ \beta > 0$, where α is the shape parameter and β is the rate parameter. Its density is $p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda}$, $\lambda > 0$. The normalizing constant $\frac{\beta^\alpha}{\Gamma(\alpha)}$ does not depend on λ, so $p(\lambda) \propto \lambda^{\alpha-1}e^{-\beta\lambda}$.
3. By Bayes' theorem, $p(\lambda \mid y) = \frac{p(y|\lambda)p(\lambda)}{p(y)}$. The denominator is $p(y) = \int_0^\infty p(y \mid \lambda)p(\lambda)\, d\lambda$. Since $p(y)$ does not depend on λ, $p(\lambda \mid y) \propto p(y \mid \lambda)p(\lambda)$.
4. Substitute the Poisson likelihood and gamma prior: $p(\lambda \mid y) \propto \left[e^{-\lambda}\lambda^y\right]\left[\lambda^{\alpha-1}e^{-\beta\lambda}\right]$. Collect powers of λ and exponential terms: $p(\lambda \mid y) \propto \lambda^{y+\alpha-1}e^{-(\beta+1)\lambda}$. Equivalently, $p(\lambda \mid y) \propto \lambda^{\alpha+y-1}e^{-(\beta+1)\lambda}$.
5. Recall that the kernel of a gamma distribution with shape a and rate b is $\lambda^{a-1}e^{-b\lambda}$. Comparing, $a - 1 = \alpha + y - 1$ so $a = \alpha + y$. Also, $b = \beta + 1$. Therefore, $\lambda \mid y \sim \text{Gamma}(\alpha + y, \beta + 1)$.
6. To verify the full posterior density, include the normalizing constant: $p(\lambda \mid y) = \frac{(\beta+1)^{\alpha+y}}{\Gamma(\alpha+y)}\lambda^{\alpha+y-1}e^{-(\beta+1)\lambda}$, $\lambda > 0$. This is exactly the density of $\text{Gamma}(\alpha + y, \beta + 1)$.
7. Therefore, a gamma prior combined with a Poisson likelihood produces a gamma posterior. Hence, the gamma distribution is a conjugate prior for the Poisson likelihood.
8. Interpretation:

$$\alpha_{\text{post}} = \alpha + y, \quad \beta_{\text{post}} = \beta + 1.$$

The observed count y is added to the prior shape parameter, and one observation is added to the prior rate parameter.